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Challenges and strategies for globally resilient shrimp aquaculture

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Challenges and strategies for globally resilient shrimp aquaculture

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Abstract

Blue Foods from aquaculture and fishing vessels are essential for global diets and economies, yet shrimp aquaculture faces mounting challenges in supply, disease, climate, and market sustainability. This review analyzes global shrimp aquaculture systems, identifies key resource flows and sectoral stressors, and presents five resilience strategies. It emphasizes solution-driven pathways and highlights barriers that must be addressed collaboratively to ensure sustainable and resilient shrimp supply chains.

Introduction

For more than 4000 years, indigenous communities around the world have practised various types of aquacultural production. Since the 1960's, aquaculture has become industrialised, promising a sustainable solution to food security often referred to as '*The Blue Revolution'* [1]. Through this revolution, many species including aquatic animals, plants and algae known as *'Blue Foods'*, are caught, or cultivated in freshwater and marine environments [2,3]. These foods are a vital and undervalued component of global food systems and are the fastest growing food sector today [4]. Foods obtained from oceans, lakes and rivers are a linchpin to many cultures, diets, economies, and livelihoods with 10% of the Earth's population relying on Blue Foods for income with around half of these foods farmed in aquaculture rather than wild caught [3,5]. Blue Foods provide nearly 20% of global animal protein and are the main source of animal protein by volume, two times that of poultry protein and three times larger than cattle [4,5]. Beyond protein, they are critical sources of micronutrients and essential fatty acids, such as vitamin B12 and omega 3, especially important to vulnerable populations [4]. Around 72% of global aquaculture relies on feed for production. Exceptionally good Feed Conversion Ratios (FCR), which capture the efficiency of the conversion of feed to the final weight of animal biomass, indicate that protein and nutrient supply from fed aquaculture

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come at a lower environmental cost compared with most terrestrial animals [3,5–7].

Global Blue Food consumption more than doubled by edible weight between 1961 and 2020, and demand has also roughly doubled since the beginning of the 21st century, with China as the largest consumer [3]. Critical to food security, domestic aquaculture production at a national level is positively correlated to the per capital consumption of blue foods[8]. Spurred by continued urbanisation, Blue Food demand is predicted to double again by 2050 [8,9]. Despite this, a complex mix of challenges, from resource shortages to disease outbreaks, extreme weather, changing climates, and economic factors have contributed to decelerated growth rate in this sector over the last thirty years [5]. Recent literature indicates that aquaculture infrastructures and practices will fail to meet growing demand by approximately 50 million tonnes [10,11,12,13]. Current and future aquaculture industries must become more resilient to both known and unforeseen disturbances to secure the supply of safe, nutritious, and sustainable Blue Food for the world's growing population.

Today, global trade of farmed Blue Foods remains limited to a small number of species, with over 89% of aquaculture output sold to local markets [14]. Catfish, tilapia, salmon, and shrimp represent one-third of the value of international seafood trade, but only 8% of global Blue Food aquaculture [14,15]. All four Blue Foods are produced in response to differing social, economic, and environmental drivers, guided by a wide range of regulatory landscapes managing challenges such as food safety, fraud, and environmental stewardship [16]. While catfish and tilapia are farmed largely for domestic markets in developing countries, salmonid aquaculture targets international markets in more developed countries. Uniquely, shrimp aquaculture caters for both domestic and international markets, with a great degree of variation in production processes. Reflecting on the global importance of shrimp as a major source of protein, this review focuses on shrimp as an exemplar Blue Food, farmed for both international trade markets and domestic consumption.
This Review, provides an overview of global shrimp farming systems, explores four overlapping thematic challenges impacting the sector and presents five strategies to build more resilient and responsible shrimp aquaculture industries.

2 Global shrimp consumption and aquaculture

Shrimp products are supplied in a wide range of forms: live, fresh, frozen, raw or partially cooked, deveined, deheaded, pre-peeled, pre-seasoned or included in processed goods such as ready meals. Options vary depending on cultural preferences, price and point of consumption and reflect the quality of farm inputs and outputs. Approximately 30% of shrimp consumed comes from wild capture, comprising numerous species that are generally lower in value by weight [17]. In contrast, farmed shrimp supply 70% of the global market, with production heavily concentrated on two species, Black Tiger (*Penaeus monodon*) and Pacific Whiteleg (*Litopenaeus vannamei*), which together represent 80% of aquaculture output [17].

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Figure 1. A global heat map of the regions where the majority of shrimp production takes place. Data for this was extracted from the FAO 2024 report which detailed production levels in 2024. (FAO, 2024)

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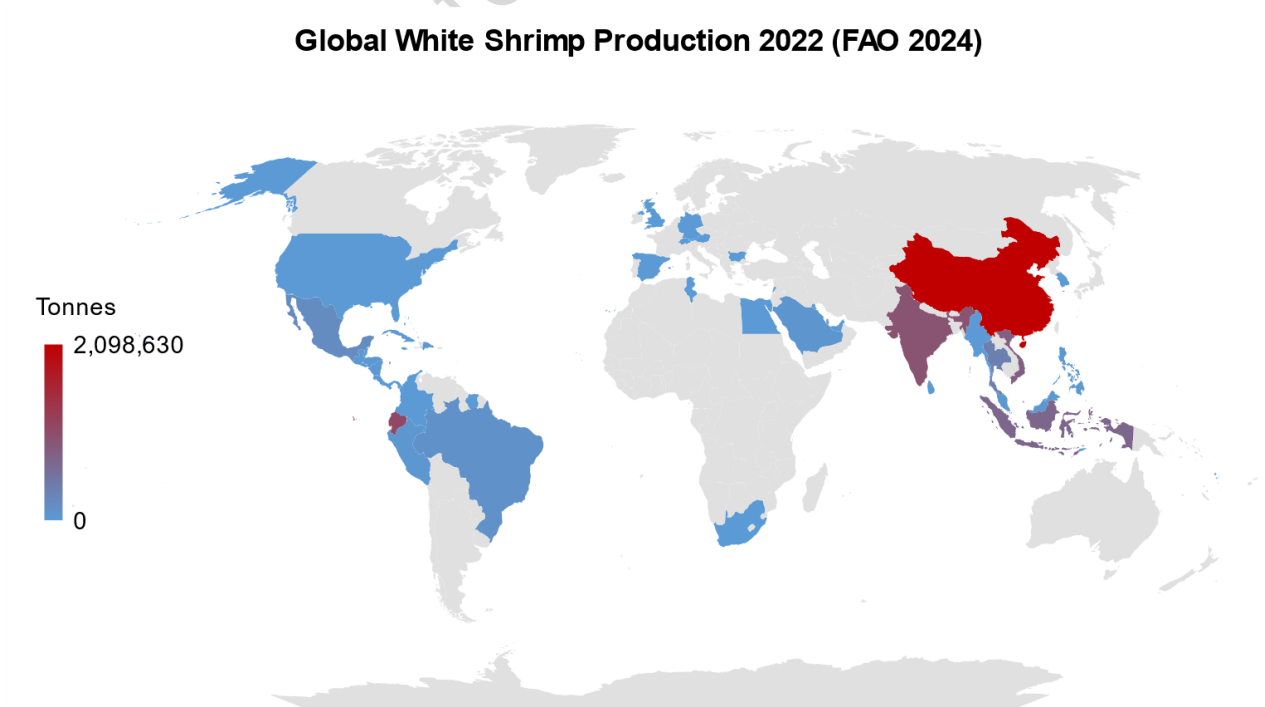

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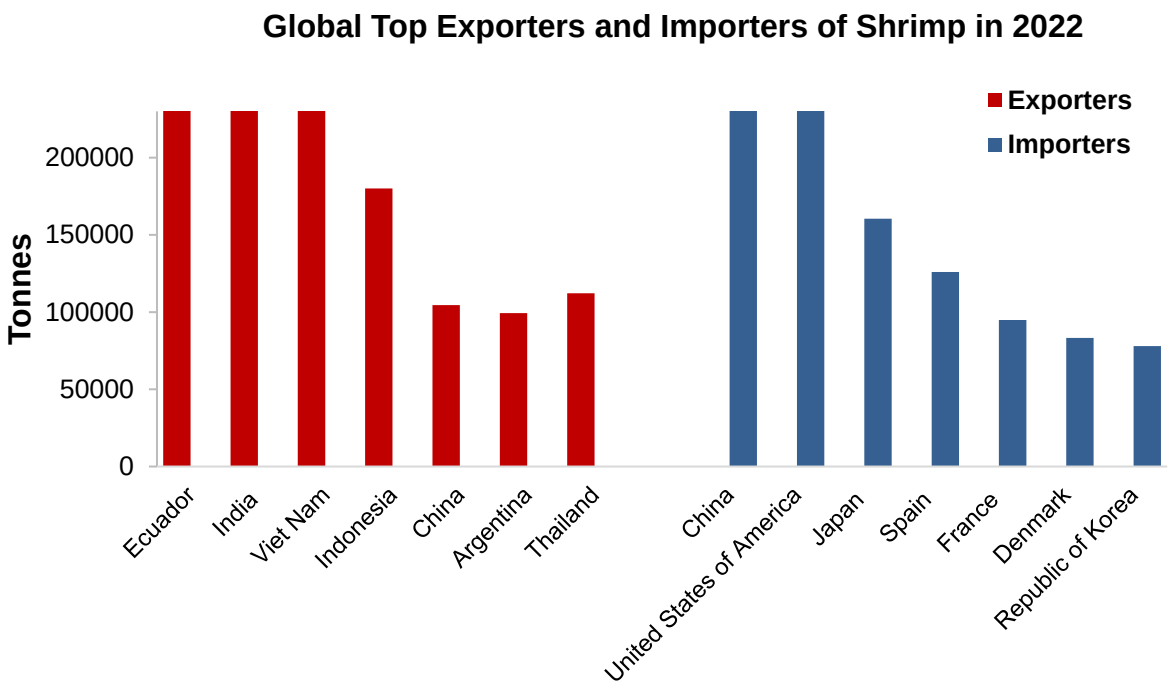
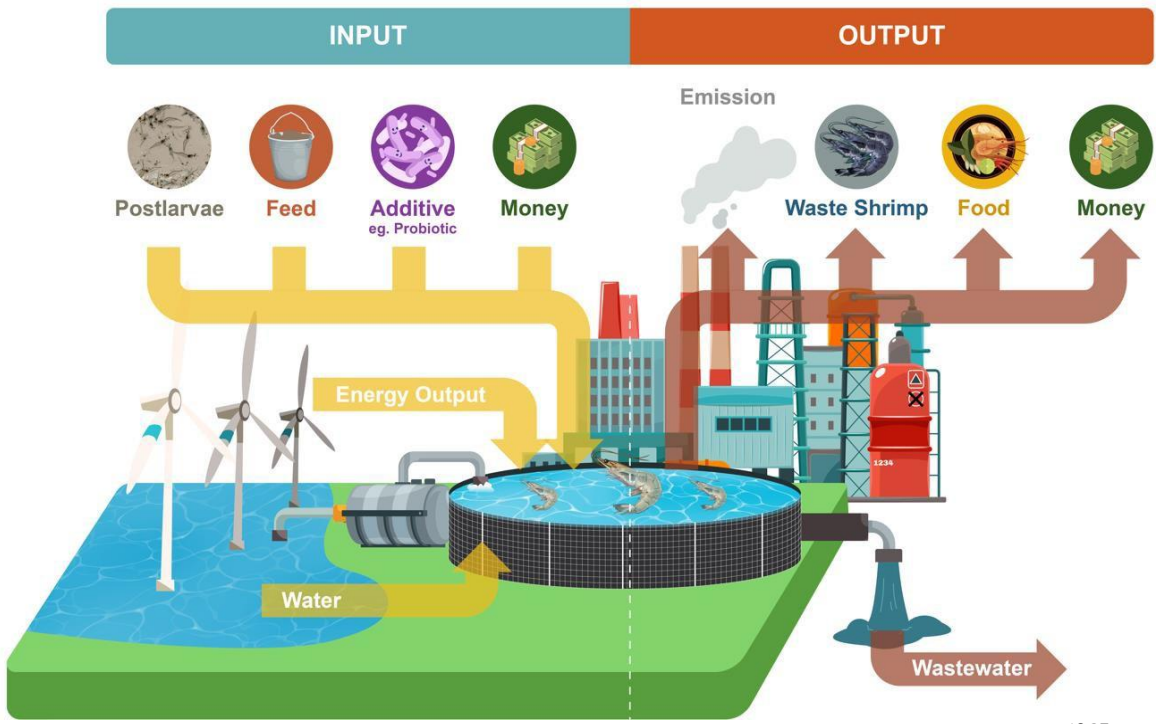


Figure 2. The scale of exports and imports of shrimp globally in 2022 (FAO, 2024).

Native to Southeast Asia, *P. monodon* was a mainstream product until the late 1990s when outbreaks of yellow head virus and white spot syndrome virus catalysed a shift in production species. Today, Pacific Whiteleg is 52% of world shrimp aquaculture, with around 5.8 million tonnes produced in 2020 [20,21]. Pacific Whiteleg shrimp is believed to be superior in large farms with higher volumes of production and better FCRs. This species has a shorter culture time of between 60 and 90 days, while Black tiger shrimp normally takes over 100 days [22,23]. This means that Pacific Whiteleg shrimp aquaculture comes with lower economic risk if crops fail due to disease or environmental distress. Targeting international markets, Black Tiger shrimp is recognised as a premium product due to its colour, taste, texture, and bigger size, while lack of availability may also be a factor. This species tends to perform better in smaller, less intensive farming systems [21].

Today, shrimp farming is concentrated in Latin American and East and Southeast Asia with most supply flowing to high-income markets in North America, Europe, and Japan [24,25]. Farming approaches are contingent on a mix of factors, from cultural norms to geo-politics, labour costs, disease prevalence, trade relationships and availability of capital. Both extensive and semi-intensive systems tend to be run by independent smallholders for domestic markets. In contrast, the increased volume of production, as well as high capital and operational costs afforded by intensive and super-intensive systems means that production tends to focus on international export, financially supported by large industry actors. Some of the giants in the global shrimp production sector are Industrial Pesquera Santa Priscila (Ecuador), Thai Union Group PCL (Thailand), Apex Frozen Foods Ltd (India), Admiralty Island Fisheries Inc. (USA), Grand Ocean Seafoods Co. Ltd (Taiwan), Charoen Pokphand Foods PCL (Thailand), Ocean More Foods Co. Ltd (China), Seaprimexco Vietnam (Vietnam), Mazzetta Company LLC (USA), Liveris Afentoulis & Co. (Greece), The Waterbase Ltd (India), Grupo Ibérica Congelados S.A. (Spain), Nordic Seafood A/S (Denmark), Pacific Seafood (USA), Zeal Aqua Ltd (India), and Nissui Corporation (Japan). Other than these prominent leaders in global shrimp production, there is an emergence of startups to satisfy increasing shrimp demand or to provide value-added services for shrimp farmers and exporters, such as Jala (Indonesia), XpertSea (Canada), AquaConnect (India), Delos Aqua (Indonesia), Shiok Meats (Singapore), Tepbac (Vietnam),

AquaExchange (India), Atarraya Inc (Mexico), eFishery (Indonesia), Eva Keferböck (Austria), and Aquapurna (Germany). Some of them are also involved with new global trends and cutting-edge technologies such as machine learning, artificial intelligence, the Internet of Things, automation, cell culture technology, Recirculatory Aquaculture System (RAS), and co-culture technology. The widely growing global shrimp industry is now also significantly influenced by sustainability focus, adequate certifications for more consumer acceptance, varying export standards, post-pandemic market fluctuations, genetic improvement, value addition, and product innovation, blockchain technology for traceability, alternative feed sources, and more research and development activities to supply rising food requirements in more sustainable and economic [24]. Typically, shrimp aquaculture relies on the supply of several key inputs: larvae and post-larvae, feed, additives, energy, and water. Historically, larvae were caught in open water, but with growing concerns around disease as well as damage to and reduced availability of wild stock, more are being grown in aquaculture [25,26]. There is an undersupply of certified disease-free larvae now, and they can cost up to six times more than non-certified options. Around 70% of greenhouse gas emissions and at least



50% of production costs are attributed to the use of commercial feed in intensive shrimp farming [18,27]. Efforts have been made to reduce the number of wild fish incorporated into feeds. Now only 10% of fishmeal in feed is from wild caught and fish waste sources, despite this, shrimp remains the second largest consumer of wild fish in aquaculture, after salmon [18,28]. Water quality is fundamental to shrimp aquaculture, and applying feed and other additives to it, including probiotics, disinfectant, or insecticide, causes farm management challenges. Though just outside the focus of this paper, it is worth noting that shrimp shells are a valuable key output and waste product for shrimp processors. Around 50-60% of post-harvest shrimp is inedible, but the chitin in shrimp shells can be used in a range of industries, from biotechnology, biochemistry, and pharmaceuticals, to paper, food, and the textiles industry – for purposes such as food preservation, bioplastic production, nitrogen doping, and wastewater treatment [29,30]. For example, recent research has used chitin for macro and micronutrient rich additive for shrimp feed production and bio-fertiliser in urban aquaponic systems [26–28].

Figure 3. Key inputs and outputs of the global shrimp supply chain, focusing on the shrimp farm level. Each has been identified as having major role in the long term economic and environmental sustainability of the shrimp aquaculture sector.

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Shrimp aquaculture occupies a significant land footprint, around 2.4 million hectares worldwide [31]. Land use in shrimp farming varies depending on the productive system used and there are generally understood to be four categories of shrimp aquaculture systems: extensive, semi-intensive, intensive, and super-intensive [32]. Extensive systems use more land while intensive systems use more embodied land, often linked to feed production.

For semi-intensive, intensive, and super-intensive systems, shrimp is produced in two stages which can take place in different growing environments. The first is the larval stage comprising of the hatching and nursing of shrimp eggs to produce post-larvae (PL). In the second stage, post-larvae are grown out to mature shrimp. Some farms specialise in different stages while others support the full growth cycle, for example, in extensive grow-out production systems PL is grown in large tidal ponds until it is harvested for consumption. **Table 1** provides a breakdown of the typical characteristics of each production system, focusing on the post-larvae stage of production for Pacific White leg shrimp.

Extensive systems tend to have low operational costs. Aeration and water pumping are limited and there is no need to add feed. While there are examples of high productivity in extensive grow out systems, most tend to have low stocking densities, producing 1-2 crops per year with lower stocking density. These systems have a direct relationship with coastal landscapes, which give rise to environmental concerns around land and resource use, water pollution, and destruction of ecologically important aquatic zones such as mangroves [33]. Regular discharge and consumption of fresh or brackish water mean that these systems come with a high-water footprint. Policy changes to tackle dwindling reserves of global freshwater threaten these water consumption practices. The study conducted considering 150 extensive shrimp farms in Bangladesh revealed that farmers have an advantage of low-lying tidal land which grants water circulation through natural tidal variations but also have an inadequacy of acquiring more land area to extend their farms [34]. Semi-intensive aquaculture is characterized by slightly higher stocking densities, the use of artificial feed and water pumping. Intensive systems require more resource inputs and procedures, like pond lining and washing between crops. In these systems, land and resource consumption tends to be more efficient but more economically and environmentally costly [21 31]. Regular water exchange is required to manage the breakdown of organic waste bi-products. While water recirculation reduces water consumption, this has onward impacts on embodied water consumption within the feed in these densely stocked ponds [28]. An analysis of the 200 farmers provided technical inefficiencies associated with their intensive white-leg shrimp farming in Vietnam and they were denoted by 22% seed inefficiency, 32% energy inefficiency, and 88% shrimp yield production. It was suggested to improve them by proper stock management, technological innovations, and using post larvae with better quality [36]. Some systems use water recirculation, negating the need for water exchanges through biological filtration, disinfection, and aeration. In these instances, technologies such as Biofloc and aquamimicry used in countries such as Brazil, Vietnam, and Thailand , use microbes to control water quality and reduce water consumption, while additional aquaculture ponds for other species can be used to handle nitrogenous waste as well as diversify production [33,37]. From a recent study on white leg shrimp production in Brazil, the intensive system performed well in terrestrial ecotoxicity, human toxicity, marine ecotoxicity, abiotic depletion, and aquatic ecotoxicity, while the super-intensive biofloc system impacted lesser for global warming, ozone layer depletion, abiotic depletion fossil fuels, eutrophication, acidification, and photochemical oxidation [38]. Moreover, from 230 white shrimp farmers in Vietnam, it was exhibited that the highest environmental efficiency and highest yield were from super-intensive farming while intensive farming resulted in the maximum nutrient surplus due to the highest feed requirement and lowest environmental efficiency [39-41]. A comparison between emergy indices of Ecuador's semi-intensive shrimp aquaculture, Brazilian extensive and super-intensive cultures revealed Brazilian super-intensive system had the largest annual yield per hectare, lowest unit emergy value (efficient resource usage), and highest renewability indicator, while the Brazilian extensive system showed the highest Emergy Yield Ratio indicating its' exploitation of local inputs. However, all systems caused Emergy Sustainability Indices (ESI) to be lower than 1

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denoting lower sustainability, and the super-intensive system comparatively had a higher ESI due to the use of biofloc technology [42-43]. Super-intensive systems come with even higher stocking densities and resource consumption. Indoor Recirculating Aquaculture Systems (RAS) and aquaponic systems are a burgeoning sub-set of this typology [44]. Super intensive systems are gaining traction, particularly in areas with limited land. Though costly to build and run, they are providing important steps towards a closed production system and opportunities to resituate production closer to urban environments to secure aquatic protein supply and reduce logistics.

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Table 1: Characteristics, inputs, and outputs of Pacific Whiteleg shrimp aquaculture systems [32,35]

Production System	Extensive	Semi-Intensive	Intensive	Super-Intensive
Characteristics				
PL Pond Size	0.5 – 20 ha pond	0.5 – 20 ha pond Intertidal zone	0.25 – 1.0 ha lined pond embankment	0.25 – 0.5 ha fully lined pond
PL Pond Location	Intertidal zone		Water reservoir and canals	
Stocking Density	Low (< 5 PL/ m^2)	Medium (5 – 15 PL/ m^2)	High (20-100 PL/ m^2)	High (>100 PL per m^2)
Production cycle	4-6 months	3-4 months	2-3 months	2-3 months
Yield/ crop	200 – 500 kg/ ha (1-2 crops per year)	1,500 – 2,000 kg/ ha (<3 crops per year)	3,000-10,000 kg/ ha (3-4 crops per year)	20,000 kg/ ha (3-4) crops per year)
Inputs and Outputs				
PL source	Wild	Hatchery (certified and non-certified)		
Feed	Natural feed	Natural and artificial feed	Artificial feed	Artificial feed and shrimp may feed on bacterial flocs
Pre/Probiotics used?	Yes	Yes	Yes	Yes
Fertiliser used?	May be used to enhance natural production		Fertilizer may be used to enhance natural production at the beginning	No
Disinfectant used?	No		Yes	
Water aeration?	No	Yes		
Management of waste in water	Natural breakdown by microorganisms			Biofloc system
Energy source	Regionally specific. Most use grid electricity with back-up diesel generators. Some use solar power.			

3 Challenges to resilient and responsible shrimp aquaculture

Shrimp aquaculture faces a range of resilience challenges. We explore four complex and overlapping challenge themes touching on food security, food safety, climate change, resources, pollution, trade, investment, and oversight.

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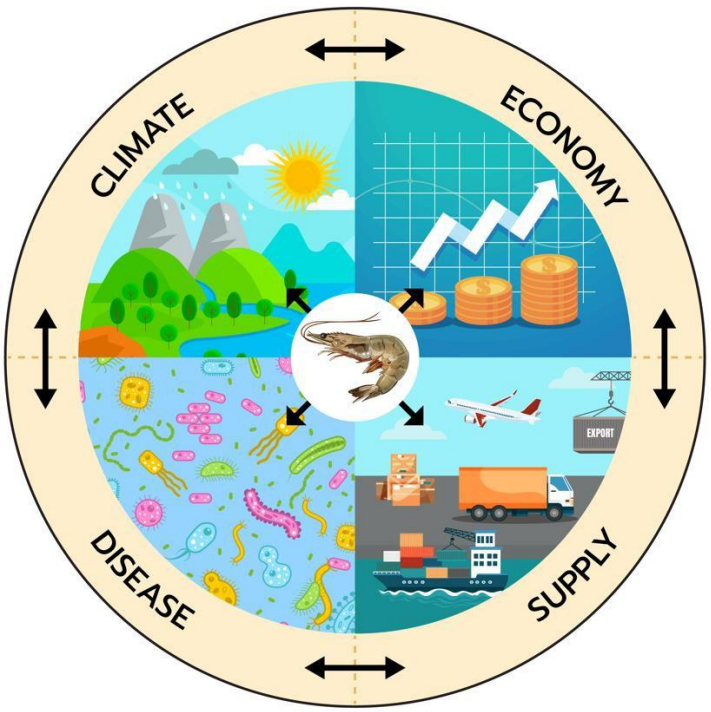

Figure 4. A representation of the four overlapping challenges identified affecting the resilience of the global shrimp aquaculture sector.

3.1 *Supply*

Intensive shrimp aquaculture relies on a steady supply of inputs such as feed and additives. Supply of broods s tock, larvae, or post-larvae are all equally critical. These supply chains also rely on primary raw material supply. In the case of feed, raw material inputs include fish meal, often from wild caught fish, wheat bran, probiotics, and vitamins. Supply chains for these direct and indirect inputs are complex and increasingly vulnerable to political and climate events **(3.3)**. As aquaculture practices expand in developing countries, so too does competition for these ingredients, with knock-on impacts on global trade agreements, price fluctuations and taxation **(3.4)**. Regional expansion of farms also places substantial strains on water supply by fuelling competition with other forms of agriculture and settlements. This can become more critical in instances of drought or when water quality is compromised due to disease, bacteria overgrowth, or pollution **(3.2, 3.3).** The same can be said for the vulnerability of energy supply for pond aeration where breaks in grid supply can result in almost immediate and financially catastrophic crop loss.

Maintaining safety and integrity is an additional challenge along complex shrimp supply chains. Increased distance between industry actors from farm and fork can reduce transparency and traceability, in turn increasing risks to food safety and fraud. Global food safety alerts captured by the EU's Rapid Alert System for Food and Feed (RASFF) highlight a range of recurring issues with shrimp including high levels of allergens such as sulphur dioxide, fraudulent health certificates, extensive use of antibiotics, spoilage linked to poor temperature control, and a range of pathogens like Listeria and Salmonella [45] **(3.2)**. Especially during the past few years, the COVID-19 pandemic also caused issues in shrimp feed with respect to fluctuating demands, supply chain disruptions, rising freight costs, and border restrictions [46-48]. Other than that, shrimp feed cost is also currently influenced by the Russia–Ukraine war, rising shipping rates, and extended droughts [48]. Despite the feed supply challenges, the growth of the shrimp feed industry can be seen with a focus on producing affordable, sustainable, and readily available feed materials. Some prominent leaders in shrimp feed are CP Foods (Thailand), Guangdong Haid (China), Skretting (Norway), Cargill (USA), BioMar (Denmark), ADM (USA), and Avanti Feeds Ltd (India). For instance, the largest

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manufacturer and exporter of shrimp feed in India, Avanti Feeds Ltd recently achieved a 53.2% year-on-year increase corresponding to a net profit of 1.1365 billion Indian rupees (~$13.5 million) [46-48].

3.2 Disease

Shrimp pathogens spread rapidly causing significant mortality epidemics. Major diseases impacting production in the past three decades include Acute hepatopancreatic necrosis disease, white spot syndrome virus and yellow head virus infections, all are linked to high mortality [49]. While these diseases can vary in their impact and prevalence in different shrimp farming regions and species, they can cause massive financial losses **(3.4)** [52] There are susceptible to a wide range of diseases as their immune responses are not adaptive and rely mainly on innate immunity to defend against pathogens [49]. Once a pond is infected disease poses another massive and growing threat to global shrimp production. Disease outbreaks can be caused by poor farm management practices with a lack of water quality monitoring, gaps in biosecurity, and poor shrimp nutrition. Shrimp pathogens can be spread within or amongst farms through the movement of infected stocks, vectors (e.g. birds, rodents) and fomites (e.g. people, equipment, people). Little can be done in terms of mitigation when disease outbreaks occur. Perera et al. showed that shrimps from Northwestern and Eastern provinces of Sri Lanka had a high white spot syndrome virus infection rate (≥83 %) which was greater in broodstock, and growing shrimp compared to post-larvae and also revealed the genetic diversity and adaptability of the virus [53]. Currently, there are some initiatives and implementations proceeded by the shrimp and biotech industries to address these diseases; such as an advanced technology (Shrimp MultiPath2.0) for early detection of diseases by Genics (Australia), edible RNA substance with novel encapsulation delivery to cope with diseases by Skretting (Norvey) and ViAqua (Israel), an information system to detect, monitor, and report diseases by eFishery (Indonesia), and a water conditioning solution (AQUALISAN) by Solvay (Belgium). Facing huge economic losses **(3.4),** farmers may turn to the use of antibiotics or decide to harvest earlier to reduce crop losses, but both strategies can lead to potential adverse impacts on human health **(3.1).** If high rates of mortality do occur, this incurs further work and costs to, for example, clean ponds and purchase new inputs to begin growing again. Often, farmers may struggle to afford such actions. Improper cleaning between crops can lead to subsequent disease outbreaks, while improper disposal of diseased shrimp could lead to disease outbreaks between farms and spill-over to natural aquatic ecosystems **(3.3).** Disease prevention is by far the preferred and optimum approach. To prevent disease epidemics, farmers face further challenges including the cost of implementation, lack of policy incentives, the need for joined-up action at local and regional scales, and gaps in education and training in terms of improving on-farm biosecurity.

3.3 *Environment*

Shrimp aquaculture is also threatened by a range of climatic factors including global warming, changing precipitation patterns, sea level rise associated with tropical cyclones, acidification, cryosphere melting, and greater frequency and intensity of extreme weather events [64]. Anthropogenic carbon emissions cause ocean warming, acidification, and oxygen loss, with further impacts on water quality, from bioaccumulation of organic pollutants, waterborne pathogens, and algal blooms, impacting ecosystem and human health [55,56]. For example, extreme wave heights contributing to sea level changes, coastal erosion and flooding have increased 1.0 cm per year [57], and both paleoclimate and modern observations suggest that the strongest El Niño and La Niña events since the pre-industrial period have occurred during the last fifty years. Despite the intensification of extreme weather, the expansion of coastal shrimp aquaculture, especially tidal pond-based production systems, have been a primary driver for mangrove deforestation. This deforestation leads to a reduction in protection of coastlines and coastal habitats [58]. Though deforestation has slowed considerably in the last twenty years; without mangroves, coastal populations remain vulnerable to extreme weather events while ecosystems lack a stable environment to thrive [59]. This vulnerability has had an indirect impact on the expansion of the shrimp farming sector and has increased competition for and the cost of land next to water bodies that are also protected from extreme weather **(3.4)** [16].

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Climate change also impacts shrimp health and disease prevalence. Shrimp immunity becomes compromised with spikes in temperature and salinity, and shrimp pathogens thrive in warmer water. Temperature spikes can accelerate pathogen growth, shorten incubation time, and increase the severity of disease. Extreme weather events such as hurricanes, cyclones, and heavy rains can also cause changes in water temperature, salinity, and nutrient levels, creating ideal conditions for large disease outbreaks. Changes in levels of precipitation or drought make it difficult for farmers to maintain water quality and salinity levels to prevent further outbreaks **(3.2)**.

The environmental challenges outlined are not evenly distributed. Countries that contribute least to the drivers of climate change are often the most vulnerable to it. For example, the largest climate impacts are experienced by developing and Least Developed Countries (LDCs), and island populations, small-scale food producers, low-income households, and Indigenous Peoples [60]. When climate impacts halt the supply of inputs or cause disease impacts, this may present more economic challenges to smaller, independent farms serving domestic markets than larger farms backed by large corporations that produce shrimp for international markets **(3.1, 3.2, 3.4).**

3.4 *Economics*

Export-focused shrimp farms tend to be of the intensive or super-intensive variety, with financial support coming from shrimp intermediaries. These intermediaries can provide access to funds for capital costs and a steady stream of income; however challenges arise when shrimp crops are lost. The stakes are high, especially if farmers are in debt and cannot fulfil supply arrangements. For example, the average debt of farming households within Thailand in 2022 was estimated at over 400,000 THB ($14,000) with the mean number of loans per household being 3.42 and over half of those loans from 'informal' institutions [61]. Lenders are often situated in local communities and act as intermediaries in relation to the sales, transport, and processing of the harvest. They have significant influence on what farms produce and how farmers source and invest which can undermine social and environmental factors [62,63].

Demand is steadier through these relationships, but farms are often required to attain international certifications, such as through the Best Aquaculture Practices (BAP) scheme to meet customer needs. While these standards aim to assure production is sustainable and safe, farms require significant investment to continue to meet these standards. This can lock-in financial relationships and limit farmer agency in the long-term. Another knock-on effect of these certification schemes is that they can drive unethical behaviours and fraud to meet international standards [64]. In developing countries, aquaculture industries for domestic consumption tended to have lower environmental and economic sustainability performance indicators compared to export products, suggesting that pressure from international markets could help drive higher environmental standards [8]. Smaller, independent, operations for domestic markets are more often extensive or semi-intensive. They are generally not certified due to the associated costs and low potential return on investment. These producers tend to have less access to financial support to refine operations towards economic resilience.

Shrimp aquaculture can be financially very risky, with the chance of losing whole crops due to disease outbreaks or adverse weather **(3.2,3.3).** The negative impact of viral diseases in shrimp aquaculture is so large that there are extended times of price increases globally as production growth cannot keep up with demand [52] Fluctuation in the price of key inputs, especially feed, further threatens the economic stability of operations **(3.1).** The investment required to reduce crop loss is often out of reach for smaller farms, meaning that their operations are at further financial risk. When crops are lost, this can have a devastating impact on many livelihoods. This tendency of 'boom and bust' on farms is also seen at a global scale. The cost of labour, availability of land and other resources, climate and disease events can tip the scales on whether whole countries can remain competitive on the global market. This is compounded further by tax agreements with large importers like the European Union and, in some instances, trade embargoes

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stemming from concerns around welfare and environmental damage.

4 Resilience Strategies

While many solutions to build globally resilient shrimp aquaculture already exist, there is not one clear outlier. A matrix of solutions across scales, sectors and timeframes may be more appropriate. Reflecting all five stages of the resilience cycle: Prepare, Prevent, Protect, Respond and Recover; existing solutions in science and technology, business, policy, and finance solutions have been reviewed and clustered under five distinct but interlinked thematic strategies: Detect, Insulate, Streamline, Diversify and Support [65-66].

4.1 Detect

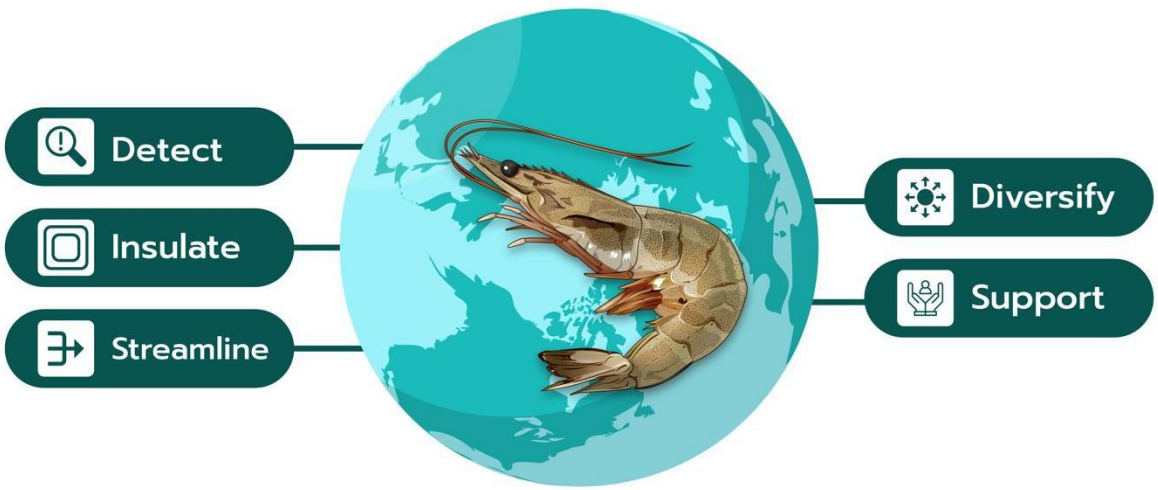

Figure 5. Five different resilience strategies have been identified to support the economic and environmental future shrimp aquaculture sector.

Various technologies are available to enable farms to detect operational threats. Uptake of these technologies could support farmers to take action to prevent or protect against the impact of threats and subsequent financial loss. Early-warning systems may be used to forecast extreme weather events, communicating both risk and potential actions to mitigate impact. For example, risk of an air temperature spike in the short-term could prompt farmers to increase oxygenation to improve crop resilience. In the medium to long-term, sustained environmental monitoring and forecasting may inform policies on where to best situate shrimp production based on current and future data predictions. The likelihood of disease outbreaks can be predicted or detected at different scales. In ponds, continuous monitoring of pond water quality, for example, oxygen, salinity, and microbial communities, can signal farmers to address issues before disease outbreaks occur, such as increasing oxygenation and water exchanges. Rapid pond-side diagnostics can also detect changes in farmed shrimp DNA and the presence of pathogens, providing real-time data on shrimp health and early indicators of disease, which can help farmers decide when to harvest [67]. At regional levels, such technologies can detect larger disease outbreaks to predict at-risk zones or onward impacts, like a shortage of post-larvae or the need for prebiotics. Shrimp pathogens can be tracked along supply chains through traceability processes that record data on health and origin at various stages of production, processing, retail, and consumption [68]. Sustained patterns in historical data and disease trends can be used to detect pathogens early, and identify sources; make data and artificial intelligence (AI) driven management decisions; enhance biosecurity; assure product quality, build consumer confidence; and support continuous improvement efforts. Beyond disease, supply

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chain traceability can also enable the detection of food fraud and spoilage while highlighting potential points of failure linked to extreme weather, geo-political conflict, raw material shortages or increased consumer demand. On farms, this may lead to preventative efforts to, for example, source feeds from different suppliers with better levels of traceability. For consumers, detection may be understood more broadly as demand for more transparency, to make purchasing decisions based on a range of data, from food safety to nutrition, labour rights, animal welfare, resource use and biodiversity impacts. Much work is required to develop and maintain such datasets for consumer-centric detection, but rapid progression in digital technologies such as blockchain and AI offer future promise in this area [69-70].

4.2 Insulate

Biosecurity measures are vital to insulate against disease outbreaks, these include: controlled access to farms, the use of more disease-resistant strains, sourcing health certified post-larvae, the use of prebiotics, the installation of nets over ponds to keep out wild birds/rodents and implementing disinfection practices such as foot dipping to prevent outbreaks between ponds and farms. Farms may also use feeds designed to enhance the gut microbiome of shrimp to improve digestion and boost immunity [71-72]. Shading devices over ponds and reservoirs to buffer water scarcity and excess rainfall can provide additional insulation against extreme weather [73]. Solar power would enable farmers to insulate against unreliable grid electricity and unregulated costs of diesel generation while providing opportunities for additional income from the sale of excess energy. Nature-based solutions could provide further valuable insulation effects to both disease, climate, and economic threats. Planting mangroves within ponds can create natural shading in extreme heat while encouraging stress-reducing, health-enabling aquatic ecosystems. Mangroves can reduce the intensity of coastal storms and flooding, and restoration of natural ecosystems along coastlines, can further insulate the shrimp sector, particularly farms that are situated near coastlines [71]. Polyculture production, where more than one species is grown together, can have similar benefits to manage water quality while providing more than one source of income. Insurance covering climate, disease, and supply threats would provide a much-needed economic buffer for farmers [74]. For the latter threat, dependable food supply is of increasing importance, particularly for small, remote, island and densely populated contexts. For example, the small city-state of Singapore imports almost 90% of its food [75]. As a coastal city keen to localise protein supply, the Singapore Government aims to increase local seafood production yield from current levels of 8% to 30% by 2030 [76]. Intensive urban aquaculture will play a critical role in solutions that increase the resilience of food supply by bringing food production closer to the point of consumption. Several land-based vertical and off-shore aquaculture systems are in development, with ongoing challenges around the cost of the former and water quality for the latter [77,78]. In its infancy, these hyper-local solutions come with high capital and operational costs. But with lower logistics costs and less steps between producer and consumer they may become more viable if supply chains continue to become less reliable.

4.4 Streamline

Actions to achieve resource efficiency are already practiced in larger, more intensive aquaculture production facilities, with plenty of lessons for smaller resource-constrained farms. Deploying incremental steps to reduce resource consumption and waste requires investment but through policies and financial backing, there are opportunities to trickle existing approaches down to smaller farms. Improving FCRs and reducing the variation in FCRs found, especially in small holdings, in terms of reducing the major cost input is of utmost importance [5,79]. Reusing water through recirculation is another good practice. Similarly, use of expensive additives and energy for oxygenation may be limited through improved biosecurity, and monitoring for anti-nutrients in feed, especially mycotoxins [80]. By streamlining production systems to international best practices, confirmed through independent and regulated certification schemes such as Good Aquaculture Practices (GAP), farms can be more prepared to tackle stressors when they arise, as well as improve the overall sustainability of operations. These schemes outline responsible farm practices, such as optimum stocking densities and feeding regimes, to reduce stress factors and bolster

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disease and climate resilience. By complying with such schemes, farmers are better placed to sell into international markets thus securing and improving incomes. To do so, farmers need economic and policy support as well as agreed codes of practice joined up across whole supply chains. For example, retailers could incentivize this shift by setting criteria for certification in product specifications and mandate the use of selected certification schemes in food production and imports, support relevant research and development and enable farmers to access soft loans for the implementation of new technologies and infrastructures. Collaboration between key actors is key to streamlining across the shrimp sector. If achieved, this would be a key driver towards transparent and traceable supply chains. Regional collaborations such as the Asian Seafood Improvement Collaboration (ASIC) bring together producers, processors, environmental NGOs, local certification bodies and export market stakeholders to develop innovative tools and agree standards to improve regional production. Further work of this nature is needed to establish networks of cooperation between international agencies while ensuring that small-scale farms have a voice and agency in these efforts.

4.5 Diversify

Unreliable supply chains, changing climates, difficult to reach sustainability targets, disease outbreaks and fluctuating costs and thus profits or indeed, in some cases, losses, point towards the need to diversify farm inputs and outputs. Depending on the shrimp species and their preferred water salinity, farmers may adapt operations to diversify production through co-cropping with different aquatic species within ponds. This can produce cheaper, live feedstock for young shrimp larvae such as Artemia, which grow well in salty conditions, or produce plants through appropriate irrigation systems [82]. There is also potential to further reduce the consumption of wild caught fishmeal and reduce the carbon footprint of production through diversification of inputs. And novel feeds from alternative protein sources (plants, insects, and algae), tailored to the shrimp microbiome, present major opportunities for the sector [14,15]. For example, some feeds made from insect meal have been shown to improve growth and the immunity of shrimp [81]. However, shifts to diversify feed sourcing may well have undesirable consequences that need consideration. Plant-based options for feed in intensively farmed monogastric animals, such as poultry and pigs, are often soy-based and have been linked to deforestation. A whole sector shift to such a plant-based feed would increase competition for other terrestrial animal feed, particularly poultry, which could accelerate deforestation and increase costs through increased competition for supply.

4.6 Support

Underpinning all resilience strategies is the need for a variety of support mechanisms to make meaningful and impactful change. While many potential solutions exist, it is particularly difficult for smallholders to change how they operate due to the lack of access to capital. Shrimp farmers, especially smallholders, need appropriate economic advice and support to bolster resilience to the many food-system shocks they are facing. Micro-finance organisations could also provide an alternative to the existing informal financial institutions to fund necessary improvements [82]. Likewise, reputable cooperatives could support clusters of small farms to pool funds to bulk buy resources, localise the production of raw materials, and share monitoring technologies within communities. Connecting smallholders with more industry-leading farms would serve to facilitate technical and financial knowledge exchange. Increasing levels of debt will only hinder progress, therefore policies must incentivise sustainable economic support. For example, in 2023, in response to having Asia's highest levels of household debt, Thailand's cabinet approved a plan to suspend debt repayments for three years for millions of farmers, with farmers who have kept up with debt repayments allowed to borrow up to another 100,000 baht ($2800 USD) from the state-owned Bank for Agriculture and Agricultural Co-operatives [83]. It will also be vital for governments and industry to take a holistic lens in planning future investments. For example, funding targeted research and development could help to translate existing practices to small, less well financed farming practices with due consideration of social, economic, and environmental impacts.

5 Future Outlook

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The resilience strategies proposed in the previous section provide a framework to consider how different but interconnected solutions might address the multiple production challenges outlined. These strategies cannot operate as silos but require a matrix-based approach, which recognizes the overlapping relationships between them. For example, streamlining aquaculture processes through certification schemes has onward benefits to detection of disease outbreaks along supply chains. Likewise, through pathogen detection, farms can insulate their income by harvesting shrimp early. There are substantial barriers to implementation of the strategies and solutions highlighted. Much of the change needed is contingent on top-down action, through policy support, increased access to sustainable funding mechanisms and implementation-led research and development. Shifts to more resilient and responsible aquaculture approaches must be incentivized, benefiting and empowering farmers as a result. Focusing on increasing farm productivity alone will not meet the challenges to feed the world sustainably by 2050. To secure sustainable future shrimp demand, these shifts require an ecosystem of perspectives towards join-up action between key actors across the entire supply chain, from the feed producers to the supermarkets and food service industries. Combining the mix of the solutions discussed could reshape how shrimp farms are equipped, managed, funded, and monitored, which would inform newly adapted, place-specific models for shrimp aquaculture. Further work is required to understand how such solutions might be incentivized and implemented in different regions, particularly across a huge network of smaller, resource-constrained farms. Alongside government and policy makers, NGOs and shrimp farming suppliers, processors and retailers, cross-disciplinary networks could support farmers to co-create a suite of solutions that are grounded and actionable in the immediate and medium term, spanning local and global scales. These bottom-up and top-down networks could be truly game-changing in reimagining a resilient future for global shrimp farming with vital lessons to be learned from and others to be applied to other aquaculture systems.

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Author contributions

Conceptualization: EC, NK, AP, WR, CE; Methodology: EC, JB, AP, CE; Investigation: EC, PB, LC, NK, AP, WR, CE; Writing – original draft: EC, JB, PB, BB, SC, WNC, LC, NK, KM, AP, WR, CE; Writing – review & editing: EC, JB, WNC, LC, NK, AP, WR, CE; Supervision: CE; Project administration: EC; Funding acquisition: CE.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Competing Interests

Awanwee Petchkongkaew (AP) served as an Associate Editor in 2024 and had no involvement in the peer review process of this manuscript. Similarly, Chris Elliott (CE), who has served as Associate Editor or Co-Editor-in-Chief since 2017, also had no role in the peer review process of this manuscript.

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Figure legend

Figure 1. A global heat map of the regions where the majority of shrimp production takes place. Data for this was extracted from the FAO 2024 report which detailed production levels in 2024. (FAO, 2024)

Figure 2. The scale of exports and imports of shrimp globally in 2024 (FAO, 2024).

Figure 3. Key inputs and outputs of the global shrimp supply chain, focusing on the shrimp farm level.

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Each has been identified as having major role in the long term economic and environmental sustainability of the shrimp aquaculture sector.

Figure 4. A representation of the four overlapping challenges identified affecting the resilience of the global shrimp aquaculture sector.

Figure 5. Five different resilience strategies have been identified to support the economic and environmental future shrimp aquaculture sector.

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